

INFORMATION TECHNOLOGY

Regulation : 2021
Semester: 04
Subject code & Subject Name : 21CS401 DESIGN AND ANALYSIS OF ALGORITHMS
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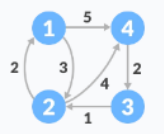
UNIT 1

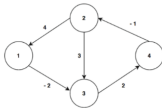
Marks	Sno	Question	Level	Co Mapped
0.5/1	1	Algorithm analysis aims to evaluate the _____ of an algorithm.	Remember	CO1
	2	True or False: Space Complexity refers to the amount of space an algorithm requires to store data during its execution.	Remember	CO1
	3	Time Complexity in algorithm analysis measures: a) The number of instructions an algorithm executes. b) The space an algorithm occupies in memory. c) The accuracy of an algorithm's output. d) The input size of the algorithm.	Remember	CO1
	4	In algorithm efficiency analysis, which of the following is NOT a factor considered? a) Time Complexity b) Space Complexity c) Code readability d) Input size	Remember	CO1
	5	True or False: Asymptotic notations like Big O, Big Omega, and Big Theta are used to describe the growth rate of an algorithm's	Remember	CO1
	6	Recursive algorithms break down problems into: a) Larger subproblems b) Smaller subproblems c) Similar problems d) Unrelated problems	Remember	CO1
	7	Divide and Conquer is a problem-solving technique that divides a problem into _____ subproblems.	Remember	CO1
	8	True or False: Big O notation provides an upper bound on an algorithm's time complexity.	Remember	CO1
	9	Dynamic Programming is particularly useful for solving problems with: a) No overlapping subproblems b) Overlapping subproblems c) Recursive subproblems d) Limited input size	Remember	CO1
	10	Asymptotic notations like Big O, Big Omega, and Big Theta are used to describe the _____ behavior of algorithms.	Remember	CO1
	11	Non-recursive algorithms are often preferred for their simplicity and _____ space requirements.	Remember	CO1
	12	True or False: Polynomial time complexity indicates that the algorithm's running time increases linearly with input size.	Remember	CO1
	13	Big Theta notation represents: a) Lower bound b) Upper bound c) Exact bound d) None of the above	Remember	CO1
	14	True or False: Algorithm analysis considers only time complexity and ignores space complexity.	Remember	CO1
	15	The primary goal of mathematical analysis in algorithm design is to: a) Evaluate code readability b) Assess algorithm correctness c) Predict algorithm efficiency	Remember	CO1
2	16	Define recursion.	Remember	CO1
	17	How do you measure the efficiency of an Algorithm?What is basic operation?	Understand	CO1
	18	What is an Algorithm?	Understand	CO1
	19	Define recurrence relation.	Remember	CO1
	20	Write down the properties of an asymptotic notations.	Remember	CO1
	21	Give the Euclid's Algorithm for computing gcd(m,n)	Create	CO1

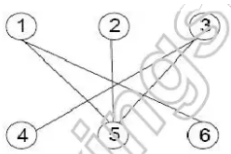
	22	Define big theta notations	Remember	CO1
	23	Define big omega notations	Remember	CO1
	24	What is the order of growth?	Understand	CO1
	25	Write the recursive algorithm for computing the fibonacci series.	Understand	CO1
14,7	26	i) Explain Big O notations in detail with suitable examples.(7) ii) Describe the properties of algorithm and define the procedure to write an algorithm.(7)	Understand	CO1
	27	Write an algorithm to perform insertion sort and derive its time complexity.	Create	CO1
	28	Write an algorithm to perform binary search on a sorted array. Find its best, worst and average case complexities.	Create	CO1
	29	a) List out the steps involved in non- recursive algorithm with an example b) Explain string matching algorithm and calculate the time complexity (7)	Understand	CO1
	30	i Discuss the steps in Mathematical analysis for non recursive algorithms. Do the same for finding the factorial of a number.(7) ii Solve the following recurrence equation using backward substitution and tree method: $T(n) = T(n/2) + 1$, where $n = 2^k$ for all $k \geq 0$ (7)	Apply	CO1
	31	Give the general plan for analyzing the time efficiency of recursive algorithms and use recurrence to find number of moves for Towers of Hanoi problem	Create	CO1
	32	a)Design an algorithm to find whether all the elements in a given array are distinct and find the time complexity. (7) b) Use brute force method to find closest pair of points and calculate the time complexity	Create	CO1
	33	Analyze the Insertion sort algorithm and find its Best, Average and Worst case	Analyze	CO1
	34	Write an algorithm for Greatest Common Divisor.	Create	CO1
	35	i)Explain the fundamentals of problem solving (7). ii)Explain the mathematical analysis of analysis of non-recursive algorithm(7).	Understand	CO1
	36	Explain the general framework for analysing the efficiency of algorithm.	Understand	CO1
	37	Discuss about the recursive and non-recursive algorithm with example.	Understand	CO1
	38	Discuss the Best,Worst and Average case analysis of an algorithm with example.	Understand	CO1
	39	Explain asymptotic notations in detail with suitable examples.	Understand	CO1
10	40	Develop a recursive algorithm to sort 'n' numbers using quick sort. Solve its recurrence relation using substitution method.	Create	CO1
	41	Prove the time complexity of the matrix multiplication is $O(n^3)$	Evaluate	CO1
	42	What is the general plan for time efficiency of recursive algorithm? Give an example	Remember	CO1
	43	Find the solution for KnapSack problem. When $n=3$, $(W1,W2,W3)=(18,15,10)$. $(P1,P2,P3)=(25,24,15)$ and $M=15$ using Brute Force.	Apply	CO1
	44	Write an algorithm for linear search.Derive the best,average and worst case analysis.	Create	CO1
	45	Discuss the time and space complexity of performing a linked list operation.	Understand	CO1
	46	Explain the important problem types of the algorithm	Understand	CO1
UNIT 2				
	1	True or False: Strassen's matrix multiplication algorithm is faster than traditional matrix multiplication algorithms for all input sizes	Remember	CO2
	2	The _____ algorithm guarantees an $O(n \log n)$ time complexity for sorting arrays.	Remember	CO2
	3	True or False: The Traveling Salesman Problem can be solved optimally in polynomial time for all instances.	Remember	CO2
	4	Heap Sort is based on the data structure called: a) Queue b) Stack c) Binary tree d) Hash table	Remember	CO2

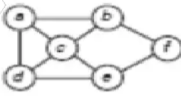
0.5/1	5	Which sorting algorithm has the worst-case time complexity of $O(n^2)$? a) Merge Sort b) Quick Sort c) Bubble Sort d) Insertion Sort	Remember	CO2																									
	6	True or False: The closest-pair problem can be efficiently solved using a divide and conquer approach.	Remember	CO2																									
	7	The Convex-Hull problem involves finding the smallest _____ enclosing a set of points.	Remember	CO2																									
	8	In the Knapsack Problem, items are selected to maximize _____.	Remember	CO2																									
	9	Brute Force string matching utilizes _____ to find a pattern within a text	Remember	CO2																									
	10	True or False: The objective of the Traveling Salesman Problem is to maximize the number of cities visited.	Remember	CO2																									
	11	The Assignment problem is concerned with finding the most efficient _____.	Remember	CO2																									
	12	. True or False: The Master Theorem is applicable to analyze the time complexity of all types of algorithms.	Remember	CO2																									
	13	True or False: Quick Sort is an example of a divide and conquer algorithm.	Remember	CO2																									
	14	True or False: The Traveling Salesman Problem can be solved optimally in polynomial time for all instances.	Remember	CO2																									
	15	Binary search is efficient for searching in _____ arrays.	Remember	CO2																									
	2	16	State the convex Hull problem.	Remember	CO2																								
		17	What is the time and space complexity of merger sort?	Understand	CO2																								
		18	What is exhaustive search?	Understand	CO2																								
19		State Masters theorem.	Remember	CO2																									
20		What is Closest pair problem?	Understand	CO2																									
21		Give the General plan for Divide and conquer Algorithms.	Analyze	CO2																									
22		Write the advantages of insertion sort.	Analyze	CO2																									
23		Derive the complexity of Binary Search.	Evaluate	CO2																									
24		What is Binary Search?	Understand	CO2																									
25		Write about traveling Salesperson problem.	Understand	CO2																									
14	26	Explain the merge sort algorithm with example	Understand	CO2																									
	27		Understand	CO2																									
	28	i)Write down the algorithm to construct a convex hull based on the divide and conquer strategy.(7) ii)Explain about Quick sort algorithm(7).	Create	CO2																									
	29	Find the Optimal solution using Branch and Bound for the following assignment problem <table><tr><td></td><td>JOB 1</td><td>JOB 2</td><td>JOB 3</td><td>JOB 4</td></tr><tr><td>A</td><td>10</td><td>5</td><td>9</td><td>5</td></tr><tr><td>B</td><td>4</td><td>4</td><td>3</td><td>7</td></tr><tr><td>C</td><td>5</td><td>7</td><td>2</td><td>8</td></tr><tr><td>D</td><td>8</td><td>6</td><td>9</td><td>4</td></tr></table>		JOB 1	JOB 2	JOB 3	JOB 4	A	10	5	9	5	B	4	4	3	7	C	5	7	2	8	D	8	6	9	4	Create	CO2
		JOB 1	JOB 2	JOB 3	JOB 4																								
	A	10	5	9	5																								
	B	4	4	3	7																								
C	5	7	2	8																									
D	8	6	9	4																									
30	State and explain matrix-chain multiplication problem. Develop an algorithm to compute an optimal order for multiplying a chain of matrices.	Remember	CO2																										
31	Explain the heap sort algorithm and sort the following integers using heap sort 10,45,80,34,67,28,53,60,19,6,23,71	Apply	CO2																										
32	(i) Explain the convex hull problem and the solution involved behind it (7) (ii) Explain the brute force method to find the two closest points in a set of n points in k dimensional space. (7)	Understand	CO2																										
	33	Explain the quick sort algorithm sort the following list of values.Write the recursive quick sort algorithm 10,45,80,34,67,28,53,60,19,23,71	Apply	CO2																									

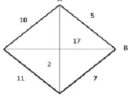
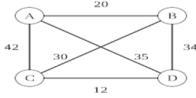
	34	What is convex hull problem? Explain the brute force approach to solve convex-hull with an example.Derive time complexity.	Understand	CO2																									
	35	Explain the String Matching algorithm using brute force approach.	Understand	CO2																									
	36	Find the optimal solution using exhaustive search method for the assignment problem given below. Persons Jobs <table><tr><th></th><th>J₁</th><th>J₂</th><th>J₃</th><th>J₄</th></tr><tr><th>a</th><td>9</td><td>2</td><td>7</td><td>8</td></tr><tr><th>b</th><td>6</td><td>4</td><td>3</td><td>7</td></tr><tr><th>c</th><td>5</td><td>8</td><td>1</td><td>3</td></tr><tr><th>d</th><td>7</td><td>6</td><td>9</td><td>4</td></tr></table>		J ₁	J ₂	J ₃	J ₄	a	9	2	7	8	b	6	4	3	7	c	5	8	1	3	d	7	6	9	4	Apply	CO2
	J ₁	J ₂	J ₃	J ₄																									
a	9	2	7	8																									
b	6	4	3	7																									
c	5	8	1	3																									
d	7	6	9	4																									
	37	i)Differentitate Exhaustive search and brute force approach(7). ii)Define Brute force approach with example(7).	Analyze,remember	CO2																									
10	38	State and explain matrix multiplication problem. Develop an algorithm to compute an optimal order for multiplying a matrices.	Remember	CO2																									
	39	Compare the stabilities and the time complexities of quicksort and the mergesort algorithm.	Analyze	CO2																									
	40	Write the algorithm to perform binary search and compute its run time complexities.	Create	CO2																									
	41	What do you mean by divide and conquer Strategy?Discuss the algorithms that follows divide and conquer strategy.	Understand	CO2																									
	42	Explain in detail about the Knapsack problem.	Understand	CO2																									
	43	Find the optimal solution to the fractional knapsack problem with given data. <table><tr><td>Item</td><td>Weight</td><td>Benefit</td></tr><tr><td>A</td><td>2</td><td>60</td></tr><tr><td>B</td><td>3</td><td>75</td></tr><tr><td>C</td><td>4</td><td>90</td></tr></table>	Item	Weight	Benefit	A	2	60	B	3	75	C	4	90	Apply	CO2													
	Item	Weight	Benefit																										
A	2	60																											
B	3	75																											
C	4	90																											
44	Explain in detail about the Assignment problem.	Understand	CO2																										
UNIT 3																													
.5/1	1	What principle is fundamental to Dynamic Programming? a) Principle of Optimality b) Principle of Greediness c) Principle of Efficiency d) Principle of Divide and Conquer	Remember	CO3																									
	2	Which algorithm is used to find the shortest paths between all pairs of vertices in a weighted graph? a) Dijkstra's algorithm b) Prim's algorithm c) Floyd-Warshall's algorithm d) Bellman-Ford algorithm	Remember	CO3																									
	3	Multistage Graphs are commonly used in which field of study? a) Economics b) Computer Science c) Biology d) Linguistics	Remember	CO3																									
	4	Optimal Binary Search Trees are designed to minimize what? a) Number of nodes b) Depth of the tree c) Search time d) Space complexity	Remember	CO3																									
	5	The Knapsack Problem involves maximizing what? a) Profit b) Weight c) Volume d) Time	Remember	CO3																									
	6	True or False: The Principle of Optimality states that an optimal solution to a problem contains optimal solutions to its subprob	Remember	CO3																									
	7	Which algorithm is used to efficiently solve the Knapsack Problem with fractional weights? a) Greedy algorithm b) Dijkstra's algorithm c) Floyd-Warshall's algorithm d) Fractional Knapsack algorithm	Remember	CO3																									
	8	Floyd-Warshall's algorithm is used to find the _____paths between all pairs of vertices in a weighted graph	Remember	CO3																									
	9	Multistage Graphs are characterized by having multiple levels of _____.	Remember	CO3																									

	10	True or False: Optimal Binary Search Trees guarantee the fastest search time for all possible search keys.	Remember	CO3
	11	Which technique does the Principle of Optimality rely on to solve problems? a) Recursive approach b) Iterative approach c) Greedy approach d) Exhaustive search	Remember	CO3
	12	The Floyd-Warshall algorithm is primarily used for finding the shortest paths in _____ graphs	Remember	CO3
	13	In the Knapsack Problem, what constraint is typically imposed on the items that can be selected? a) Weight limit b) Profit limit c) Volume limit d) Time limit	Remember	CO3
	14	True or False: Optimal Binary Search Trees guarantee the minimum possible search time for a given set of search keys.	Remember	CO3
	15	What is the main advantage of using Dynamic Programming to solve problems? a) It always guarantees the optimal solution. b) It is faster than all other algorithms. c) It can efficiently solve problems with overlapping subproblems. d) It requires fewer computational resources compared to other approaches.	Remember	CO3
2	16	State the principle of optimality.	Remember	CO3
	17	Define multistage graph. Give Example.	Remember	CO3
	18	How dynamic programming is used to solve the knapsack problem?	Understand	CO3
	19	What does Floyd's Algo do?	Remember	CO3
	20	State Assignment problem	Remember	CO3
	21	Define single source shortest path problem.	Remember	CO3
	22	List out the memory functions under dynamic programming	Understand	CO3
	23	List out the advantages of Huffman's encoding	Understand	CO3
	24	Compare greedy method and dynamic programming	Analyze	CO3
	25	Define Kruskal algorithm.	Remember	CO3
7,14	26	Explain Floyd's Algorithm for all pair shortest path algorithm with example and analyze its efficiency. (14m)	Understand	CO3
	27	Describe Knapsack problem and Memory functions with example (14m)	Understand	CO3
	28	Explain the Floyd-Warshall algorithm for finding the shortest paths in a weighted graph. Discuss its time and space complexities and provide a step-by-step example of how the algorithm works.	Understand	CO3
	29	Using OBST algorithm compute w_i, r_{ij}, c_{ij} where $j=0$ to 4 for the identifier set $\{a_1, a_2, a_3, a_4\} = \{\text{end, goto, print, stop}\}$ with $P_1 = 1/20, P_2 = 1/5, P_3 = 1/10, P_4 = 1/20, Q_0 = 1/5, Q_1 = 1/10, Q_2 = 1/5, Q_3 = 1/20, Q_4 = 1/20$ using r_{ij} construct optimal binary search tree. (16)	Apply	CO3
	30	Discuss the Principle of Optimality in dynamic programming and how it is utilized to solve optimization problems efficiently. Provide an example of an optimization problem where the Principle of Optimality plays a key role	Understand	CO3
	31	Explain Floyd-Warshall algorithm using dynamic programming. Trace the algorithm for the given example. 	Apply	CO3
	32	Apply the bottom up dynamic programming algorithm to the following instance of Knapsack Problem. Capacity $W=10$ (14m) Item Weight Value 1 7 \$42 2 3 \$12 3 4 \$40 4 5 \$25	Apply	CO3
	33	i) Explain optimal binary search tree algorithm with example and analyze its efficiency (7m) ii) Explain the basic principle of dynamic programming with suitable example. (7m)	Understand	CO3

		Solve the all pair shortest path problem for the digraph with the following weight in the matrix.(14m)																											
	34	<table><tr><td></td><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>a</td><td>0</td><td>∞</td><td>∞</td><td>3</td></tr><tr><td>b</td><td>2</td><td>0</td><td>∞</td><td>∞</td></tr><tr><td>c</td><td>∞</td><td>7</td><td>0</td><td>1</td></tr><tr><td>d</td><td>6</td><td>∞</td><td>∞</td><td>0</td></tr></table>		a	b	c	d	a	0	∞	∞	3	b	2	0	∞	∞	c	∞	7	0	1	d	6	∞	∞	0	Apply	CO3
	a	b	c	d																									
a	0	∞	∞	3																									
b	2	0	∞	∞																									
c	∞	7	0	1																									
d	6	∞	∞	0																									
10	35	Solve the following using the floyds algorithm.(10m) 	Apply	CO3																									
	36	.Write an algo to construct the optimal binary search tree given the roots $r(i,j)$, $0 \leq i-j \leq n$.also prove that this could be performed	Create	CO3																									
	37	.Find the optimal solution to the fractional knapsack problem with the given data.(10m) Item weight benefit A 2 60 B 3 75 C 4 90	Apply	CO3																									
	38	What is meant by multistage graph.Explain with an example.(10m)	Understand	CO3																									
	39	Difference between the dynamic programming and divide and conquer techniques.	Analyze	CO3																									
UNIT 4																													
0.5	1	The Simplex Method is specifically designed to solve maximum flow problems.(True/False)	Remember	CO4																									
	2	Kruskal's Algorithm and Prim's Algorithm are both used to find the minimum spanning tree of a graph.(True/False)	Remember	CO4																									
	3	The Stable Marriage Problem aims to maximize the number of stable marriages.(True/False)	Remember	CO4																									
	4	Dijkstra's Algorithm guarantees finding the shortest path in a weighted graph with negative edge weights.(True/False)	Remember	CO4																									
	5	Huffman Trees are commonly used in data compression algorithms.(True/False)	Remember	CO4																									
	6	The Simplex Method is an iterative technique used to solve _____ programming problems.	Remember	CO4																									
	7	Maximum Matching in Bipartite Graphs aims to maximize the number of _____ edges.	Remember	CO4																									
	8	The Greedy Technique makes locally optimal choices in the hope of finding a _____ solution.	Remember	CO4																									
	9	Prim's Algorithm starts with a single _____ and adds edges to grow the minimum spanning tree.	Remember	CO4																									
	10	Dijkstra's Algorithm uses a _____ to keep track of the shortest path distances.	Remember	CO4																									
	11	What problem does the Coin Change Problem aim to solve? a) Maximum Flow b) Data Compression c) Minimum Coin Change d) Shortest Path	Remember	CO4																									
	12	12. Which algorithm is commonly used to find the shortest path in a weighted graph? a) Kruskal's Algorithm b) Prim's Algorithm c) Dijkstra's Algorithm d) Huffman Trees	Remember	CO4																									
	13	Prim's Algorithm and Kruskal's Algorithm both aim to find: a) Maximum Flow b) Minimum Spanning Tree c) Maximum Matching d) Shortest Path	Remember	CO4																									

	14	The Stable Marriage Problem can be solved using the Greedy Technique.(True/False)	Remember	CO4
	15	Huffman Trees are used in data compression because they guarantee the smallest possible compressed file size.(True/False)	Remember	CO4
2	16	1.State the principle of optimality.	Remember	CO4
	17	2.Define multistagegraph.Give Example.	Remember	CO4
	18	3.How dynamic programming is used to solve the knapsack problem?	Understand	CO4
	19	4.What does Floyds Algo do?	Understand	CO4
	20	5.State Assignment problem	Remember	CO4
	21	6.Define single source shortest path problem.	Remember	CO4
	22	7.List out the memory functions under dynamic programming	Understand	CO4
	23	8.List out the advantages of huffmans encoding	Understand	CO4
	24	9.compare greedy method and dynamic programming	Analyze	CO4
	25	10.define krushkal algorithm.	Remember	CO4
7,14	26	Write a procedure for huffman code.Derive the huffman code for the following set of symbols	Remember	CO4
	27	Explain the Ford Fulkerson method for solving maximum flow problem	Understand	CO4
	28	Explain the greedy approach for finding huffman codes from the given list of symbols and their respective frequencies	Understand	CO4
	29	Write a brief note on the elements of greedy strategy	Understand	CO4
	30	What is bipartite graph? Is the subset of bipartite graph is bipartite? outline with example.	Understand	CO4
10	31	Write a brief note on the elements of greedy strategy(10).	Understand	CO4
	32	write the necessary algorithms to solve a linear programming problem using SIMPLEX method(10	Create	CO4
	33	Describe the Stable Marriage Problem and its applications. Explain the Gale-Shapley algorithm step by step using a practical example to demonstrate how it solves the Stable Marriage Problem efficiently.	Understand,Create	CO4
	34	Compare and contrast the Coin Change Problem and the Minimum Spanning Tree Problem. Discuss how greedy algorithms are used to solve these problems, including the principles behind greedy choice and optimal substructure.	Analyze	CO4
	35	Apply the maximum matching algorithm to the following using bipartite graph. 	Apply	CO4
	36	Explain Dijkstra's Algorithm for finding the shortest paths in a weighted graph. Discuss its time complexity, advantages, and limitations. Provide a scenario where Dijkstra's Algorithm would be the most suitable choice for solving a problem.	Understand	CO4
UNIT 5				
	1	The N-Queens Problem can be efficiently solved using the Greedy Technique.	Remember	CO5
	2	The Assignment Problem is a type of NP-hard problem.	Remember	CO5
	3	NP-complete problems are generally easier to solve than NP-hard problems.	Remember	CO5
	4	. P, NP, NP-Complete, and NP-Hard are classes that categorize problems based on their computational complexity.	Remember	CO5
	5	The Hamiltonian Circuit Problem aims to find a cycle that visits each vertex exactly once.	Remember	CO6
	6	The N-Queens Problem involves placing _____ queens on an N×N chessboard without attacking each other.	Remember	CO6
	7	The Branch and Bound technique is used to solve optimization problems by dividing them into _____ and _____ parts.	Remember	CO6

0.5	8	The Subset Sum Problem can be solved using a _____ approach that explores all possible subsets.	Remember	CO6
	9	The Assignment Problem aims to find the most efficient _____ of resources to tasks.	Remember	CO6
	10	NP-complete problems are those that are both in NP and are _____ to every problem in NP.	Remember	CO6
	11	What technique is commonly used to solve the Travelling Salesman Problem? a) Greedy Technique b) Backtracking c) Dynamic Programming d) Branch and Bound	Remember	CO5
	12	Which problem belongs to the NP-hard class? a) Subset Sum Problem b) Assignment Problem c) Hamiltonian Circuit Problem d) Knapsack Problem	Remember	CO5
	13	Non-Deterministic Algorithms are based on: a) Deterministic logic b) Randomized algorithms c) Backtracking d) Heuristic search	Remember	CO6
	14	The Subset Sum Problem can be efficiently solved using dynamic programming.(True / False)	Remember	CO6
	15	NP-complete problems have known polynomial-time algorithms for their solutions. (True / False)	Remember	CO6
	16	State Hamiltonian Circuit problems.	Remember	CO5
	17	Define P and NP problems.	Remember	CO5
	18	Define sum of subset problem.	Remember	CO5
	19	Differentiate Feasible and Optimal Solution.	Analyze	CO6
	20	Explain promising and non-promising node.	Understand	CO5
	21	Compare backtracking and branch bound techniques.	Analyze	CO5
	22	What are tractable and non-tractable problems?	Understand	CO6
2	23	Illustrate 8-Queens problem.	Analyze	CO5
	24	Show the purpose of lower bound.	Analyze	CO5
	25	Define chromatic number of the graph.	Remember	CO6
	26	Define backtracking.	Remember	CO5
	27	Elaborate how backtracking technique can be used to solve the n-queens problem. Explain with an example.(14m) 6.write backtracking algorithm for i) the n-queens problem ii)Hamiltonian problem iii)The subset sum problem	Evaluate	CO5
	28	Solve the following instance of Knapsack Problem by branch and bound algorithm(14m) Items Weights Values A 2 \$40 B 3.14 \$50 C 1.98 \$100 D 5 \$95 E 3 \$30	Apply	CO5
	29	Write an algorithm to determine Hamiltonian cycle in a given graph using backtracking. For the following graph determine the Hamiltonian cycle.(14m) 	Apply	CO5

7,14	30	Apply the branch and bound algorithm to solve this traveling salesman problem for the following graph.(14m) 	Apply	CO6
	31	Outline the steps to find approximate solution to NP-Hard optimization problems using approximation algorithms with an example.	Evaluate	CO5
	32	Explain the approximation algorithm for the travelling salesman problem(10m)	Understand	CO5
	33	Explain Backtracking technique with an example(10m)	Understand	CO6
	34	Implement an algorithm for knapsack problem using NP-Hard approach.(	Apply	CO5
	35	Write down and explain the procedure for tackling the 8 - queen's problem using a backtracking approach.	Understand	CO5
	36	Apply Branch and Bound algorithm to solve the travelling sales man problem for 	Apply	CO6
10	37	Explain the assignment problem in branch and bound with example.(	Understand	CO6